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Centrifugal Pump.

Centrifugal pumps are classified as rotodynamic type of pumps in which a dynamic pressure is developed which enables the lifting of liquids from a lower to a higher level.

* The basic principle on which a centrifugal pump works is that — when a certain mass of liquid is made to rotate by an external force (torque), it is thrown away from the central axis of rotation and a centrifugal head is impressed which enables it to rise to a higher level

→ Now, if more liquid is constantly made available at the centre of rotation, a continuous supply of liquid at a higher level may be ensured.

→ Since in these pumps, the lifting of the liquid is due to centrifugal action, these pumps are called "Centrifugal Pump"

→ The rise in pressure head at any point of the rotating liquid is proportional to the square of tangential velocity of the liquid at that point

i.e Rise in Pressure head = $V^2/2g$ or $\frac{\omega^2 r^2}{2g}$

→ Thus, at the outlet of the impeller, where the radius is more, the rise in pressure head will be more and the liquid will be discharged at the outlet with a high pressure head.

Due to this high pressure head, the liquid can be lifted to a high level.

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Advantages of Centrifugal Pump over Reciprocating Pump

1. The discharging capacity is very much greater in centrifugal pump as compared to reciprocating pump which can handle relatively small quantity of liquid.

2. A centrifugal pump can be used to lift highly viscous liquids such as oil, muddy and sewage water, paper pulp, sugar molasses, chemicals etc. But a reciprocating pump can handle only water or less viscous liquids

3. A centrifugal pump can be operated at very high speed without any danger of separation and cavitation; but the maximum speed of a reciprocating pump is limited.

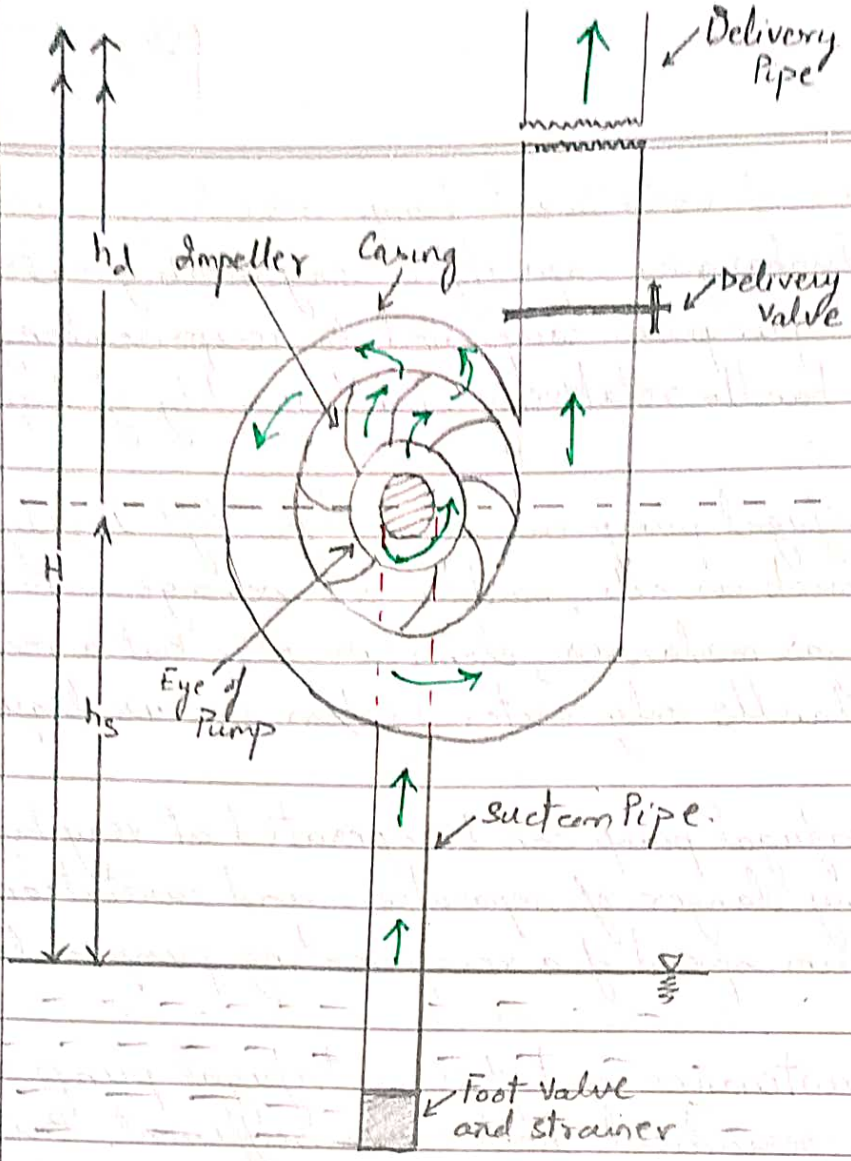
4. The maintenance cost of a centrifugal pump is low and only periodical check-up is sufficient. But for a reciprocating pump, the maintenance cost is high.

5. However, a reciprocating pump can build up very high pressures as high as 69×10^6 N/m^2 or even more. Hence these pumps are used for lifting oil from very deep oil wells.

Main Parts of a Centrifugal Pump.

The following are the main parts of a centrifugal pump :-

1. Impeller
2. Casing
3. Suction pipe with a foot valve and a strainer
4. Delivery pipe.



Impeller

The rotating part of a centrifugal pump is called 'impeller'. It consists of a series of backward curved vanes.

The impeller is mounted on a shaft which is connected to the shaft of an electric motor.

Casing

The casing of a centrifugal pump is similar to the casing of a reaction turbine.

It is an air-tight passage surrounding the impeller and is designed in such a way that the kinetic energy of the water discharged at the outlet of the impeller is converted into pressure energy before the water leaves the casing and enters the delivery pipe

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There are three types of casings.

a) Volute Casing.

→ It is a spiral type in which area of flow increases gradually.

→ The increase in area of flow decreases the velocity of flow.

→ The decrease in velocity of flow increases the pressure of the water flowing through the casing.

→ In the case of volute casing, the efficiency of the pump increases slightly as a large amount of energy is lost due to the formation of eddies.

b) Vortex Casing

→ If a circular chamber is introduced between the casing and the impeller, the casing is called vortex casing.

→ By introducing the circular chamber, the loss of energy due to formation of eddies is considerably reduced.

→ The efficiency of such a pump is more than in the case of volute casing.

c) Casing with Guide Blades.

→ The casing in which the impeller is surrounded by a series of guide blades mounted on a ring, known as diffuser. ~~is called~~

→ The guide blades are designed in such a way that the water from the impeller enters the guide vanes without shock.

→ The area of guide vanes increases, thus reducing the velocity of flow through guide vanes and consequently increasing the pressure of water.

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Suction Pipe with a Foot Valve and a Strainer

→ A suction pipe is one whose one end is connected to the inlet of the pump and other end dips into water in a sump.

→ A foot valve which is a non-return valve or one-way type of valve is fitted at the lower end of the suction pipe. The foot valve opens only in the upward direction.

→ A strainer is also fitted at the lower end of the suction pipe.

Delivery Pipe

→ A pipe whose one end is connected to the outlet of the pump and other end delivers the water at a required height is known as delivery pipe.

Working of Centrifugal Pump

→ The first step in the operation of a centrifugal pump is priming.

Priming is the operation in which the suction pipe, casing of the pump and the portion of the delivery pipe upto the delivery valve are completely filled with the liquid which is to be pumped.

→ so that all the air (or gas or vapour) from this portion of the pump is driven out and no air pocket is left.

* It has been observed that even the presence of a small air pocket in any of the portion of ~~the~~ pump may result in no delivery of liquid from the pump.

→ The necessity of priming a centrifugal pump is due to the fact that the pressure generated in a centrifugal pump impeller is directly proportional to the density of the fluid that is in contact with it.

→ But as the density of air is very low, the generated head of air in terms of equivalent metre of water head is negligible and hence the water may not be sucked from the pump.

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↝ After the pump is primed, the delivery valve is still kept closed and the electric motor is started to rotate the impeller. The delivery valve is kept closed in order to reduce the starting torque for the motor.

* The rotation of the impeller in the casing full of liquid produces a forced vortex which imparts a centrifugal head to the liquid and thus results in an increase of pressure throughout the liquid mass.

* The increase of pressure at any point is proportional to the square of the angular velocity and the distance of the point from the axis of rotation. Thus if the speed of rotation of the impeller of the pump is sufficiently high, the pressure in the liquid surrounding the impeller is considerably increased. (Now, as long as the delivery valve is closed and the impeller is rotating, it just churns the liquid in the casing.

↝ When the delivery valve is opened, the liquid is made to flow in an outward radial direction, thereby leaving the vanes of the impeller at the outer circumference with high velocity and pressure.

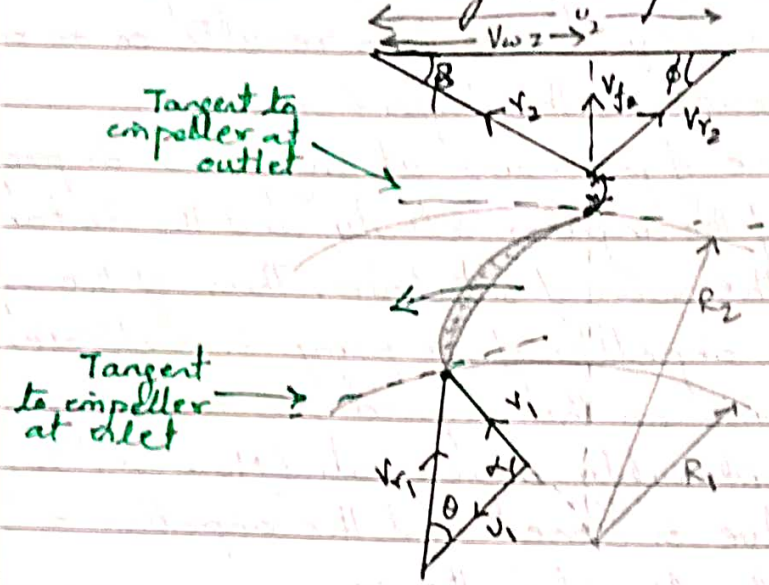
↝ At the eye of the impeller, due to centrifugal action, a partial vacuum is created. This causes the liquid from the sump (which is in atmospheric pressure) to rush through the suction pipe to the eye of the impeller, thereby replacing the liquid which is being discharged from the entire circumference of the impeller.

* The high pressure of the liquid leaving the impeller is utilized in lifting the liquid to the required height through the delivery pipe.

Work Done by Centrifugal Pump (or Impeller) on Wa

→ In case of the centrifugal pump, work is done by the impeller on the water.

→ The expression for the workdone by the impeller the water is obtained by drawing velocity triangles, a inlet and outlet of the impeller.



→ For best efficiency of the pump, the water enters the impeller radially at inlet

↳ implies absolute velocity of water at inlet mak an angle of 90° with the direction of motion of the impeller.

$\therefore \alpha = 90°$, $V_{w_1} = 0$

Same Notations

$N \rightarrow$ Speed of impeller in rpm

$D_1 \rightarrow$ Diameter of impeller at inlet

$u_1 \rightarrow$ Tangential velocity of impeller at inlet $= \frac{\pi D_1 N}{60}$

$D_2 \rightarrow$ Diameter of impeller at outlet

$u_2 \rightarrow$ Tangential velocity of impeller at outlet $= \frac{\pi D_2 N}{60}$

$V_1 \rightarrow$ Abs. velocity of water at inlet

$V_{r_1} \rightarrow$ Relative velocity of water at inlet

$\alpha \rightarrow$ ∠ b/w Abs. vel. and direct° of motion of vane at inlet

$\theta \rightarrow$ ∠ b/w rel. vel. and direct° of motion of vane at inlet

$V_2, V_{r_2}, \beta, \phi$ are the corresponding values at outlet

→ The centrifugal pump is the reverse of a radially inward flow reaction turbine.

∴ Work done by the impeller on the water per second per unit weight of water striking per second

$$= -[\text{work done in case of turbine}]$$

$$= -\frac{1}{g}[V_{w_1}u_1 - V_{w_2}u_2]$$

$$= \frac{1}{g} V_{w_2}u_2$$

Work done by impeller on water per second

$$= \frac{W}{g} V_{w_2}u_2$$

$W \rightarrow$ weight of water $= \rho g Q$

$Q \rightarrow$ Area × Velocity of flow

$$= \pi D_1 B_1 \times V_{f_1} = \pi D_2 B_2 \times V_{f_2}$$

$B_1, B_2 \rightarrow$ width of impeller at inlet and outlet

Heads and Efficiencies of a Centrifugal Pump.

1. Suction Head (h_s)

It is the vertical height of the centre line of the centrifugal pump above the water surface in the tank or sump from which water is to be lifted.

This height is also called suction lift.

2. Delivery Head (h_d)

It is the vertical height of the centreline of the centrifugal pump and the water surface in the tank to which water is delivered

3. Static Head (H_s)

The sum of suction head and delivery head is static head

$$H_s = h_s + h_d$$

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4. Manometric Head (H_m)

The manometric head is defined as the head against which a centrifugal pump has to work.

H_m = Head imparted by the impeller to the water - Loss of head in the pump

$= \frac{V_{w_2} u_2}{g}$ - Loss of head in impeller & casing

$= \frac{V_{w_2} u_2}{g}$ → if loss of pump is zero.

H_m = Total head at outlet of the pump − Total head at the inlet of the pump

$= \left(\frac{P_o}{\rho g} + \frac{V_o^2}{2g} + Z_o\right) - \left(\frac{P_i}{\rho g} + \frac{V_i^2}{2g} + Z_i\right)$

$P_o/\rho g$ → Pressure head at outlet of pump = h_d

$V_o^2/2g$ → Velocity head at outlet of pump = Velocity head in delivery pipe = $V_d^2/2g$.

Z_o → Vertical height of the outlet of the pump from datum

$P_i/\rho g$; $V_i^2/2g$; Z_i → Corresponding values at inlet i.e h_s, $V_s^2/2g$ & Z_s → at suction pipe.

$H_m = h_s + h_d + h_{f_s} + h_{f_d} + V_d^2/2g$.

h_s → Suction head

h_d → Delivery head

h_{f_s} → Frictional head loss in suction pipe.

h_{f_d} → Frictional head loss in delivery pipe.

V_d → Velocity of water in delivery pipe.

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5. Efficiencies of a Centrifugal Pump.

In case of a centrifugal pump, the power is transmitted from the shaft of the electric motor to the shaft of the pump and then to the impeller. From the impeller, the power is given to the water.

Thus power is decreasing from the shaft of the pump to the impeller and then to the water.

a) Manometric Efficiency (η_{man})

The ratio of the manometric head to the head imparted by the impeller to the water is known as manometric efficiency.

$$\eta_{man} = \frac{\text{Manometric Head}}{\text{Head imparted by impeller to water}}$$

$$\eta_{man} = \frac{H_m}{\left(\frac{V_{w_2} u_2}{g}\right)} = \frac{g H_m}{V_{w_2} u_2}$$

b) Mechanical Efficiency (η_m)

The power at the shaft of the centrifugal pump is more than the power available at the impeller of the pump. The ratio of power available at the impeller to the power at the shaft of centrifugal pump is known as mechanical efficiency.

$$\eta_m = \frac{\text{Power at the impeller}}{\text{Power at the shaft}}$$

The power at the impeller in kw $= \frac{\text{wk. done by impeller/sec}}{1000}$

$$= \frac{W}{g} \times \frac{V_{w_2} u_2}{1000}$$

$$\eta_m = \frac{W/g \left(\frac{V_{w_2} u_2}{1000}\right)}{S.P}$$

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(c) **Overall Efficiency (η_o)**

It is defined as ratio of power output of the pump to the power input to the pump.

Power output of the pump = $\frac{\text{wt. of water lifted} \times H_m}{1000}$

$= \frac{WH_m}{1000}$

Power input to the pump = S.P of the pump

$$\eta_o = \frac{WH_m/1000}{S.P}$$

\# $$\eta_o = \eta_{man} \times \eta_m$$

Q2. A centrifugal pump is to discharge 0.118 m^3/s at a speed of 1450 rpm against a head of 25 m. The impeller diameter, at outlet is 250 mm; its width at outlet is 50 mm and manometric efficiency is 75%. Determine the vane angle at the outer periphery of the impeller.

Sol. $Q = 0.118\ m^3/s$

$N = 1450$ rpm

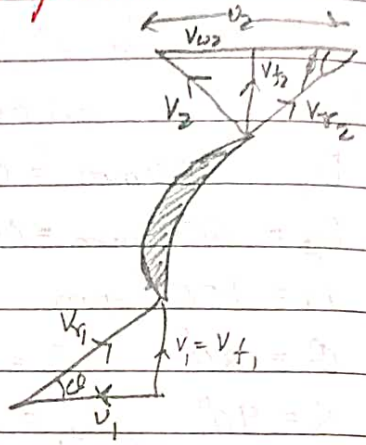
$H = 25$ m

$D_2 = 250\ mm = 0.25$ m

$B_2 = 50\ mm = 0.05$ m

$\eta_{man} = 75\% = 0.75$

Find, ϕ



$$\tan\phi = \frac{V_{f_2}}{u_2 - V_{w_2}}$$

$$\rightarrow u_2 = \frac{\pi D_2 N}{60} = 18.98\ m/s$$

$$\rightarrow V_{f_2} :- \quad Q = \pi D_2 B_2 \times V_{f_2} \Rightarrow V_{f_2} = 3\ m/s$$

$$\rightarrow V_{w_2} :- \quad \eta_{man} = \frac{g H_m}{V_{w_2} u_2} \Rightarrow V_{w_2} = 17.23\ m/s$$ (No loss)

$$\therefore \tan\phi = \frac{V_{f_2}}{u_2 - V_{w_2}} = \frac{3}{18.98 - 17.23} = 1.714$$

$$\phi = 59°44'$$

Q3. A centrifugal pump delivers water against a net head of 14.5 m and a design speed of 1000 rpm. The vanes are curved back to an angle of 30° with the periphery. The impeller diameter is 300 mm and outlet width is 50 mm. Determine the discharge of the pump if manometric efficiency is 90%

Sol. $H_m = 14.5$ m, $N = 1000$ rpm, $\phi = 30°$ (periphery → outlet), $D_2 = 0.3$ m, $B_2 = 0.05$ m, $\eta_{man} = 90\%$. Find Q.

Sol. $Q = \pi D_2 B_2 \times V_{f_2}$

V_{f_2} :- $\tan\phi = \frac{V_{f_2}}{u_2 - V_{w_2}}$

u_2 :- $u_2 = \frac{\pi D_2 N}{60}$

V_{w_2} :- $\eta_{man} = \frac{g H_m}{V_{w_2} \times u_2}$

$\rightarrow u_2 = \frac{\pi D_2 N}{60} = 15.71 \text{ m/s}$

$\rightarrow \eta_{man} = \frac{g H_m}{V_{w_2} u_2} \Rightarrow V_{w_2} = \frac{g H_m}{\eta_{man} \times u_2} = \frac{9.81 \times 14.5}{0.90 \times 15.71} = 10.06 \text{ m/s}$

$\rightarrow \tan\phi = \frac{V_{f_2}}{u_2 - V_{w_2}} \Rightarrow V_{f_2} = \tan\phi (u_2 - V_{w_2}) = 3.26 \text{ m/s}$

$\therefore Q = \pi D_2 B_2 \times V_{f_2}$

$= \pi \times 0.3 \times 0.05 \times 3.26$

$= 0.1536 \text{ m}^3/\text{s}.$

Q4. A centrifugal pump having outer diameter equal to two times the inner diameter and running at 1200 rpm works against a total head of 75 m. The velocity of flow through the impeller is constant and equal to 3 m/s. The vanes are setback at an angle of 30° at outlet. If the outer diameter of the impeller is 600 mm and width is 50 mm, determine vane angle at inlet, work done per second by impeller and manometric efficiency.

Sol. $N = 1200 \text{ rpm}$, $H = 75 \text{ m}$, $V_{f_1} = V_{f_2} = 3 \text{ m/s}$, $\phi = 40°$,

$D_2 = 0.6 \text{ m}$, $B_2 = 0.05 \text{ m}$. $D_1 = 0.3 \text{ m}$

Find, θ, wk. done, η_{man}

$u_1 = \frac{\pi D_1 N}{60} = 18.85 \text{ m/s}$; $u_2 = \frac{\pi D_2 N}{60} = 37.69 \text{ m/s}.$

$Q = \pi D_2 B_2 \times V_{f_2} = 0.2827 \text{ m}^3/\text{s}$

→ $\tan\theta = \frac{V_{f1}}{u_1} = \frac{3}{18.85}$

$\theta = 9°2'$

→ Work done by impeller on water per second.

$= \frac{W}{g} \times V_{w_2} u_2$

$= \frac{\rho g Q}{g} \times V_{w_2} u_2 = \rho Q V_{w_2} u_2$

V_{w_2} :- $\tan\phi = \frac{V_{f2}}{u_2 - V_{w_2}}$

$\tan 40° (u_2 - V_{w_2}) = V_{f_2}$

$V_{w_2} = -\left(\frac{V_{f_2}}{\tan 40}\right) + u_2$

$= 34.11 \text{ m/s}$

∴ Work done by impeller on water per second

$= \rho Q V_{w_2} u_2$

$= 1000 \times 0.2827 \times 34.11 \times 37.69$

$= 363491.28 \text{ Nm/s}$

→ Manometric Efficiency,

$\eta_{man} = \frac{gH_m}{V_{w_2} u_2}$

$= \frac{9.81 \times 75}{34.11 \times 37.69}$

$= 0.572$

$= 57.2\%$

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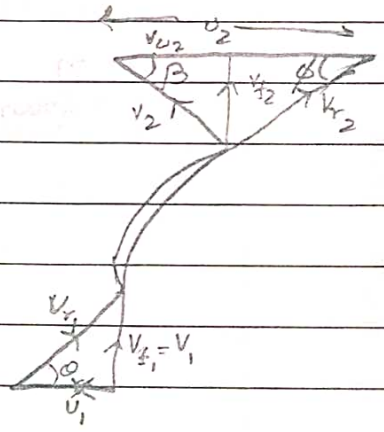
Q5. The internal and external diameter of an impeller of a centrifugal pump which is running at 1000 rpm are 200 mm and 400 mm respectively. The discharge through pump is $0.04\ m^3/s$. and velocity of flow is constant and equal to 2.0 m/s. The diameters of the suction and delivery pipes are 150 mm and 100 mm respectively and suction and delivery heads are 6m (abs.) and 30m (abs.) of water respectively. If the outer vane angle is 45° and power required to drive the pump is 16.186 kw, determine:

i) Vane angle of the impeller at inlet

ii) The overall efficiency of the pump

iii) Manometric efficiency of the pump.

Sol. $N = 1000$ rpm, $D_1 = 0.2m$, $D_2 = 0.4m$, $Q = 0.04\ m^3/s$, $V_{f_1} = V_{f_2} = 2$ m/s, $D_s = 0.15m$, $D_d = 0.1m$, $h_s = 6m$, $h_d = 30m$ (abs); $\phi = 45°$, $P = 16.188$ kw



i) $\tan\theta = V_{f_1}/u_1$ $\qquad u_1 = \frac{\pi D_1 N}{60} = 10.47$

$= 2/10.47$

$= 0.191 \Rightarrow \theta = 10°48'$

ii) $\eta_o = \frac{WP}{SP} = \frac{\frac{\rho g Q H_m}{1000}}{P}$

$= 0.02424\, H_m$

$H_m = \left(P_o/\rho g + V_o^2/2g + Z_o\right) - \left(P_i/\rho g + V_i^2/2g + Z_i\right)$

$= \left(30 + V_d^2/2g\right) - \left(6 + V_s^2/2g\right)$ $\qquad$ If $Z_o = Z_i$

V_d = Discharge / Area $= \frac{0.04}{\pi/4 \times 0.1^2} = 5.09$ m/s

V_s = Discharge / Area $= \frac{0.04}{\pi/4 \times 0.15^2} = 2.26$ m/s

$\therefore H_m = 25.06m$

$\eta_o = 0.6074 = 60.74\%$

iii) Manometric Efficiency, η_{man}

$$\eta_{man} = \frac{gH_m}{V_{w_2} U_2}$$

$$U_2 = \frac{\pi D_2 N}{60} = 20.94 \text{ m/s}$$

V_{w_2} :-

$$\tan\phi = \frac{V_{f_2}}{U_2 - V_{w_2}}$$

$$\tan 45^\circ (U_2 - V_{w_2}) = V_{f_2}$$

$$V_{w_2} = U_2 - \frac{V_{f_2}}{\tan 45^\circ} = 20.94 - \frac{2}{\tan 45}$$

$$= 18.94 \text{ m/s}.$$

$$\eta_{man} = \frac{9.81 \times 25.06}{18.94 \times 20.94} = 0.62 = 62\%$$

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Minimum Speed for Starting a Centrifugal Pump.

If the pressure rise in the impeller is more than or equal to manometric head (H_m), the centrifugal pump will start delivering water.

Otherwise, the pump will not discharge any water, although the impeller is rotating.

→ when impeller is rotating, the water in contact with the impeller is also rotating. This is the case of forced vortex.

In case of forced vortex, the centrifugal head or head due to pressure rise in the impeller

$$= \frac{\omega^2 r_2^2}{2g} - \frac{\omega^2 r_1^2}{2g}$$

ωr_1 → Tangential velocity of impeller at inlet $= u_1$
ωr_2 → Tangential velocity of impeller at outlet $= u_2$

∴ Head due to pressure rise in impeller $= \frac{u_2^2}{2g} - \frac{u_1^2}{2g}$

The flow of water will commence only if,

$$u_2^2/2g - u_1^2/2g \geq H_m$$

For minimum speed, $\boxed{u_2^2/2g - u_1^2/2g = H_m}$

Now, $\eta_{man} = \frac{g H_m}{V_{w_2} u_2}$

$$\therefore \quad u_2^2/2g - u_1^2/2g = \eta_{man} \times \frac{V_{w_2} u_2}{g}$$

Also, $u_1 = \frac{\pi D_1 N}{60}$, $u_2 = \frac{\pi D_2 N}{60}$

$$\Rightarrow \frac{1}{2g}\left(\frac{\pi D_2 N}{60}\right)^2 - \frac{1}{2g}\left(\frac{\pi D_1 N}{60}\right)^2 = \eta_{man} \times \frac{V_{w_2} \times \pi D_2 N}{g \times 60}$$

By, dividing $\frac{\pi N}{g \times 60}$ throughout.

$$\frac{\pi N D_2^2}{120} - \frac{\pi N D_1^2}{120} = \eta_{man} \times V_{w_2} \times D_2$$

$$\frac{\pi N}{120}\left[D_2^2 - D_1^2\right] = \eta_{man} \times V_{w_2} \times D_2$$

$$\boxed{N = \frac{120\, \eta_{man}\, V_{w_2}\, D_2}{\pi\left[D_2^2 - D_1^2\right]}}$$

Multistage Centrifugal ~~Pump~~ Pump

If a centrifugal pump consists of two or more impell the pump is called a multistage centrifugal pump.

The impellers may be mounted on the same shaft or on different shafts.

A multistage pump has two important functions

1. To produce a high head → Series (on same shaft)
2. To discharge a large quantity of ~~wat~~ liquid. → Parallel.

1. Multistage Centrifugal Pump for High Head

→ For developing a high head, a number of impellers are mounted in series

→ Water from suction pipe enters the 1st impeller at inlet a is discharged at outlet with increased pressure.

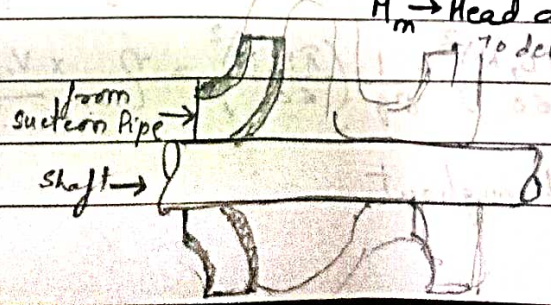
→ This increased pressure from 1st impeller is taken to the inlet of 2nd impeller.

→ At the outlet of the 2nd impeller, will now be more than that at the outlet of 1st impeller.

→ Thus, if more impellers are mounted on same shaft, the pressure at the outlet will be increased further.

Total Head developed = $n \times H_m$

H_m → Head developed by each impeller.



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2. Multistage Centrifugal Pump for High Discharge.

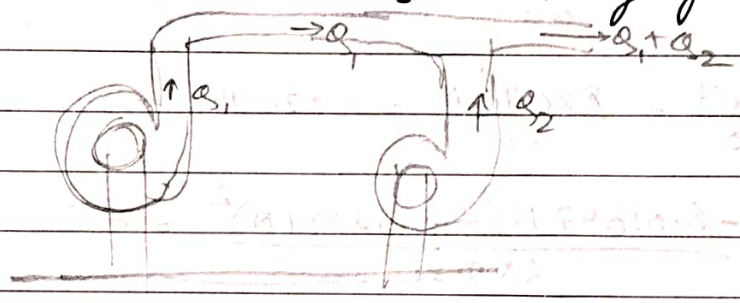
→ For obtaining high discharge, the pumps should be connected in parallel.

→ Each of the pumps lifts the water from a common sump and discharges water to a common pipe to which the delivery pipes of each pump is connected.

→ Each pump is working against the same head.

Total Discharge = $n \times Q$.

Q → Discharge from one pump.



Specific Speed of a Centrifugal Pump. (N_S)

The specific speed of a centrifugal pump is defined as the speed of a geometrically similar pump which would deliver one cubic metre of liquid per second against a head of one metre.

$$N_S = \frac{N\sqrt{Q}}{H_m^{3/4}}$$

Model Testing of Centrifugal Pumps.

1. Specific Speed Model :- $\left(\frac{N\sqrt{Q}}{H_m^{3/4}}\right)_m = \left(\frac{N\sqrt{Q}}{H_m^{3/4}}\right)_p$

2. Tangential Velocity Model :- $\left(\frac{\sqrt{H_m}}{DN}\right)_m = \left(\frac{\sqrt{H_m}}{DN}\right)_p$

3. Discharge Model :- $\left(\frac{Q}{D^3 N}\right)_m = \left(\frac{Q}{D^3 N}\right)_p$

4. Power of Pump Model :- $\left(\frac{P}{D^5 N^3}\right)_m = \left(\frac{P}{D^5 N^3}\right)_p$

Q6. The diameters of an impeller of a centrifugal pump at inlet and outlet are 20cm and 40cm respectively. Determine the minimum speed for starting the pump, if it works against a head of 25m. (2070)

Sol. $D_1 = 0.2m$, $D_2 = 0.4m$, $H_m = 25m$

Now,

$$\frac{v_2^2}{2g} - \frac{v_1^2}{2g} = H_m$$

$$v_1 = \frac{\pi D_1 N}{60} = \frac{\pi \times 0.2 \times N}{60} = 0.01047\,N \text{ m/s}$$

$$v_2 = \frac{\pi D_2 N}{60} = \frac{\pi \times 0.4 \times N}{60} = 0.02094\,N \text{ m/s}$$

$$\therefore \frac{-(0.01047N)^2 + (0.02094N)^2}{2 \times 9.81} = 25$$

$$N^2 = 1491503.901$$

$$N = 1221.27 \text{ rpm}$$

Q7. A single-stage centrifugal pump with impeller diameter of 30cm rotates at 2000rpm and lifts 3 m^3 of water per second to a height of 30m with an manometric efficiency of 75%. Find the number of stages and diameter of each impeller of a similar multistage pump to lift 5 m^3 of water per second to a height of 200m when rotating at 1500rpm.

Sol. $D_1 = 30cm = 0.3m$; $N_1 = 2000rpm$; $Q_1 = 3 m^3/s$, $H_{m_1} = 30m$,
$\eta_{man} = 75\% = 0.75$ → Single Stage Pump

→ Multistage Pump

$Q_2 = 5 m^3/s$; Total height = 200m; Height per stage = H_{m_2}
$N_2 = 1500rpm$.

The specific speed should be same

$$\therefore \left(\frac{N\sqrt{Q}}{H_m^{3/4}}\right)_1 = \left(\frac{N\sqrt{Q}}{H_m^{3/4}}\right)_2$$

$$\frac{N_1\sqrt{Q_1}}{H_{m_1}^{3/4}} = \frac{N_2\sqrt{Q_2}}{H_{m_2}^{3/4}}$$

$$\Rightarrow H_{m_2}^{3/4} = 12.411$$

$$\Rightarrow H_{m_2} = 28.74\ m$$

Hence, no. of stages = $\frac{\text{Total Head}}{\text{Head per stage}} = \frac{200}{28.74} = 6.96 \approx 7$

For diameter of impeller.

$$\frac{\sqrt{H_{m_1}}}{D_1 N_1} = \frac{\sqrt{H_{m_2}}}{D_2 N_2}$$

$$\Rightarrow D_2 = \sqrt{\frac{H_{m_2}}{H_{m_1}}} \times \frac{N_1 D_1}{N_2}$$

$$= 0.39\ m$$

$$= 39 \approx 40\ cm.$$

Q8. A four-stage centrifugal pump has four identical impellers, keyed to the same shaft. The shaft is running at 400 rpm and the total manometric head developed by the multistage pump is 40 m. The discharge through the pump is $0.2\ m^3/s$. The vanes of each impeller are having outlet angle as 45°. If the width and diameter of each impeller at outlet is 5 cm and 60 cm respectively, find the manometric efficiency.

Same shaft → so series.

Sol. No. of stages = 4, N = 400 rpm, Total manometric head = 40 m

∴ Manometric head of each stage, $H_m = 40/4 = 10$ m

$Q = 0.2\ m^3/s$; $\phi = 45°$; $B_2 = 5\ cm = 0.05\ m$; $D_2 = 60\ cm = 0.6\ m$

Manometric efficiency,

$$\eta_{man} = \frac{gH_m}{V_{w_2} u_2}$$

u_2 :- $u_2 = \frac{\pi D_2 N_2}{60} = \frac{\pi \times 0.6 \times 400}{60} = 12.56$ m/s.

V_{w_2} :-

$$\tan\phi = \frac{V_{f_2}}{u_2 - V_{w_2}}$$

$$V_{f_2} = \frac{\text{Discharge}}{\text{Area of Flow}} = \frac{0.2}{\pi D_2 \times B_2} = \frac{0.2}{\pi \times 0.6 \times 0.05} = 2.122 \text{ m/s}$$

$$\therefore V_{w_2} = u_2 - \frac{V_{f_2}}{\tan\phi}$$

$$= 12.56 - \frac{2.122}{\tan 45^\circ}$$

$$= 10.438 \text{ m/s}$$

$$\therefore \eta_{man} = \frac{g H_m}{V_{w_2} u_2} = \frac{9.81 \times 10}{10.438 \times 12.56}$$

$$= 0.7482$$

$$= 74.82\%$$

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Clevermate

Cavitation.

✓ Cavitation is defined as the phenomenon of formation of vapour bubbles of a flowing liquid in a region where the pressure of the liquid falls below its vapour pressure and the sudden collapsing of these vapour bubbles in a region of higher pressure.

→ when the vapour bubbles collapse, a very high pressure is created. The metallic surfaces, above which these vapour bubbles collapse, is subjected to these high pressures, which cause pitting action on the surface. Thus cavities are formed on the metallic surface and also considerable noise and vibrations are produced.

→ Precaution Against Cavitation.

(1) The pressure of the flowing liquid in any part of the hydraulic system should not be allowed to fall below its vapour pressure.

If the flowing liquid is water, then the absolute pressure head should not be below 2.5m of water.

(2) The special materials or coatings such as aluminium-bronze and stainless steel, which are cavitation resistant materials, should be used.

→ Effects of Cavitation

1. The metallic surfaces are damaged and cavities are formed on the surfaces.
2. Due to sudden collapse of vapour bubble, noise and vibrations are produced.
3. The efficiency of a turbine/pump decreases due to cavitation.

→ surface of blades become rough — force exerted by water on turbine blades dec. Hence, wh done by water or output HP become less and thus efficiency decreases